Automobile Parts Manufacturing Company

Marketing and Retail Analytics Project

Contents: -

- Problem statement (PART A) and Exploratory Data Analysis
- Customer Segmentation using RFM analysis
- Inferences from RFM Analysis
- Recommendations

Problem Statement (Part A)

Business Context:

- An automobile parts manufacturing company has been actively selling products to a diverse range of customers for the past three years. Despite its growth, the company lacks the in-house expertise to derive actionable insights from its transaction data. As a result, they wish to uncover hidden patterns and trends in their customer transactions. By analyzing this data, the company aims to better understand customer behavior, improve customer segmentation, and implement targeted marketing strategies. These insights will help the company not only enhance customer satisfaction but also drive revenue growth by offering more personalized and efficient services.

Objective:

The primary objective of this analysis is to leverage data science techniques to:

1. Identify underlying patterns in customer purchasing behavior.
2. Segment customers based on their transactional data.
3. Provide actionable insights to optimize the company's marketing efforts.
4. Recommend personalized marketing strategies for each customer segment to maximize sales and customer retention.

- Your role as a Business Analyst is to use the provided dataset to achieve these goals and present findings in a manner that can guide the company's decision-making.

Data Description:

- The dataset provided contains three years of transactional data from the company, with each row representing a unique order. Below is an explanation of the key attributes:
- ORDERNUMBER: Unique identifier for each order.
- QUANTITYORDERED: Number of items ordered in a specific transaction.
- PRICEEACH: Price per unit of the product in the order.
- ORDERLINENUMBER: Sequence number of the product in the order.
- SALES: Total sales value for the order.
- ORDERDATE: Date when the order was placed.
- DAYS_SINCE_LASTORDER: Number of days since the customer's previous order.
- STATUS: Current status of the order (e.g., Shipped, Disputed).
- PRODUCTLINE: Product category to which the item belongs (e.g., Motorcycles, Classic Cars).
- MSRP: Manufacturer’s Suggested Retail Price for the product.
- PRODUCTCODE: Unique identifier for the product.
- CUSTOMERNAME: Name of the customer placing the order.
- PHONE: Customer's contact phone number.
- ADDRESSLINE1: Customer's primary address.
- CITY: City of the customer's address.
- POSTALCODE: Postal code of the customer's address.
- COUNTRY: Country of the customer's address.
- CONTACTLASTNAME: Last name of the customer’s contact person.
- CONTACTFIRSTNAME: First name of the customer’s contact person.
- DEALSIZE: Size category of the transaction (e.g., Small, Medium, Large).

Part - A Exploratory Data Analysis
Univariate Analysis: -

Below are the unique values for the respective 4 attributes

Unique values of ORDERNUMBER - 298

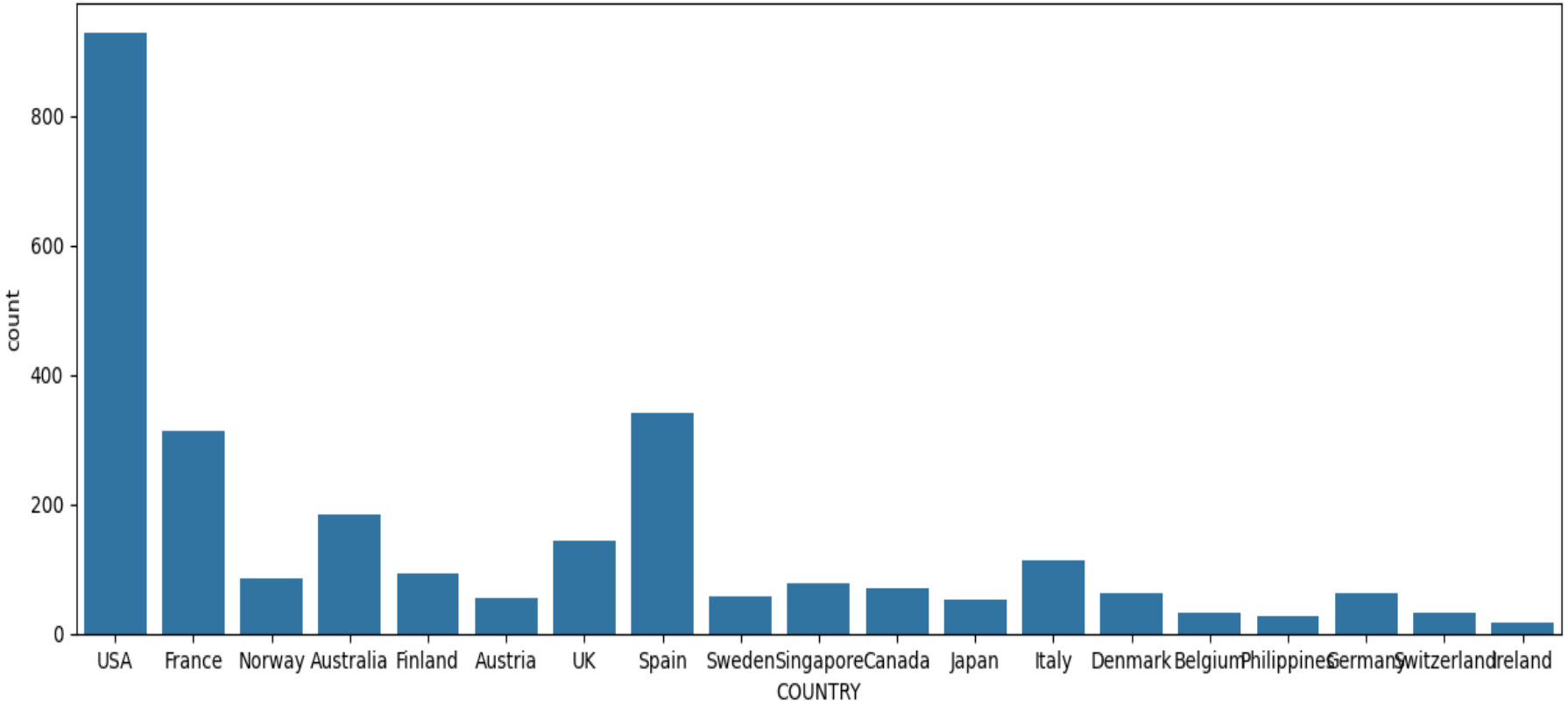
Unique values of PRODUCTCODE - 109

Unique values of CUSTOMERNAME - 89

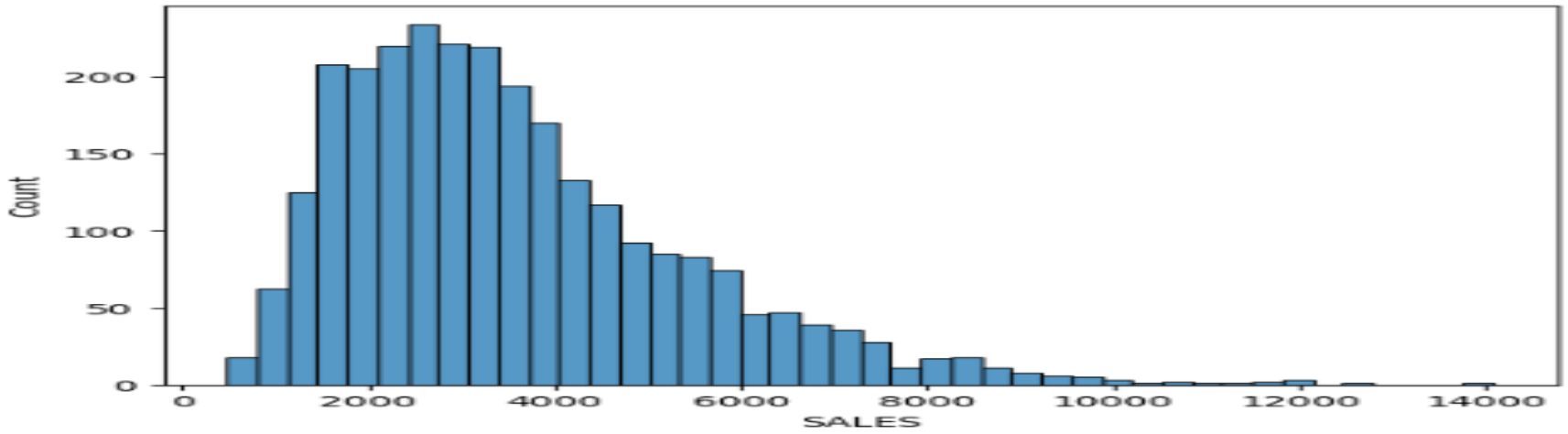
Unique values of Country - 19

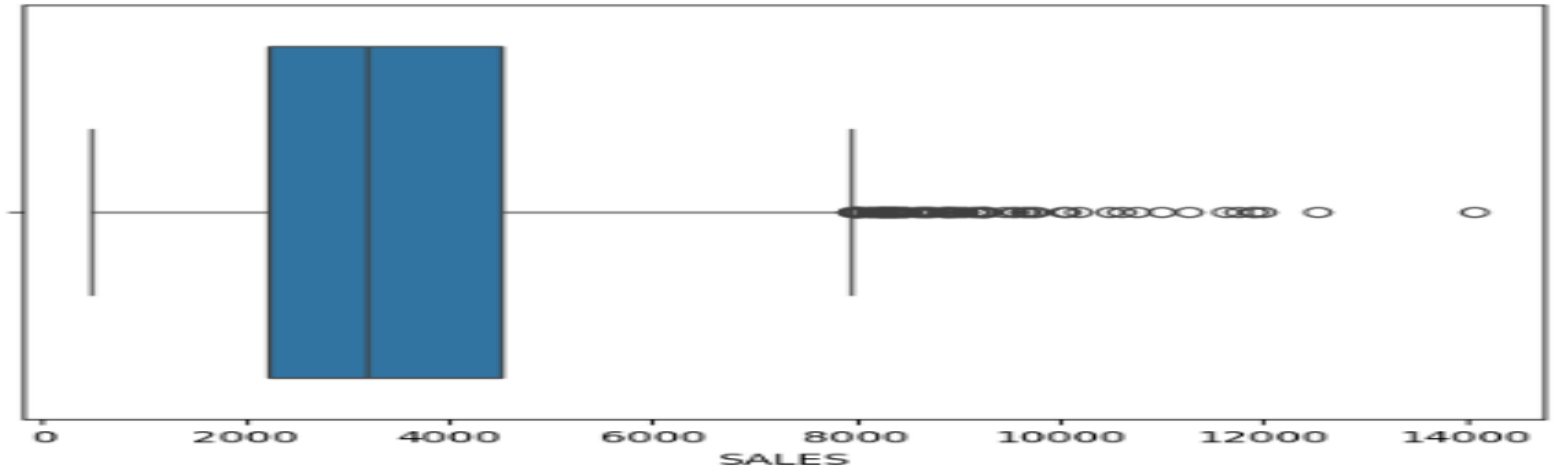
Country: -

The bar chart below shows the number of countries. USA has contributed highest portion in the market.

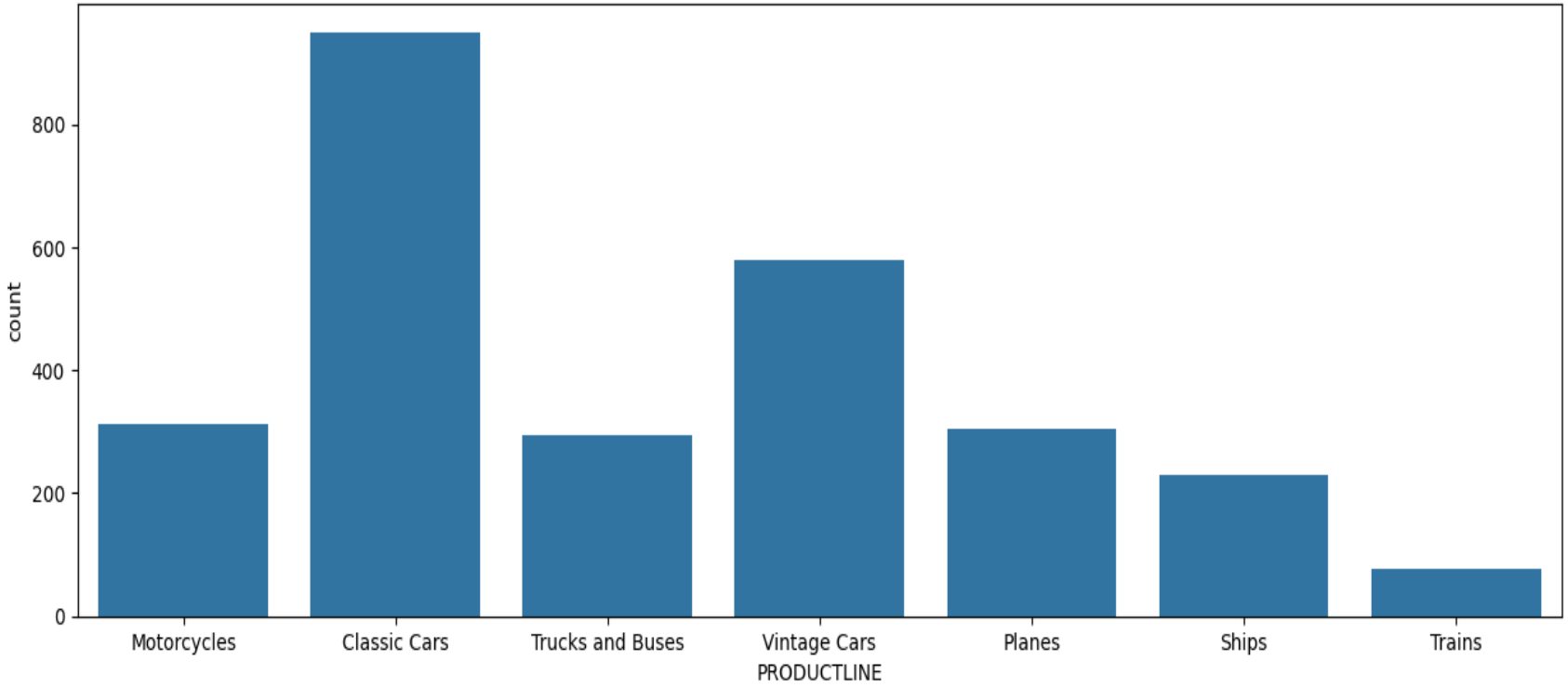


SALES of the order

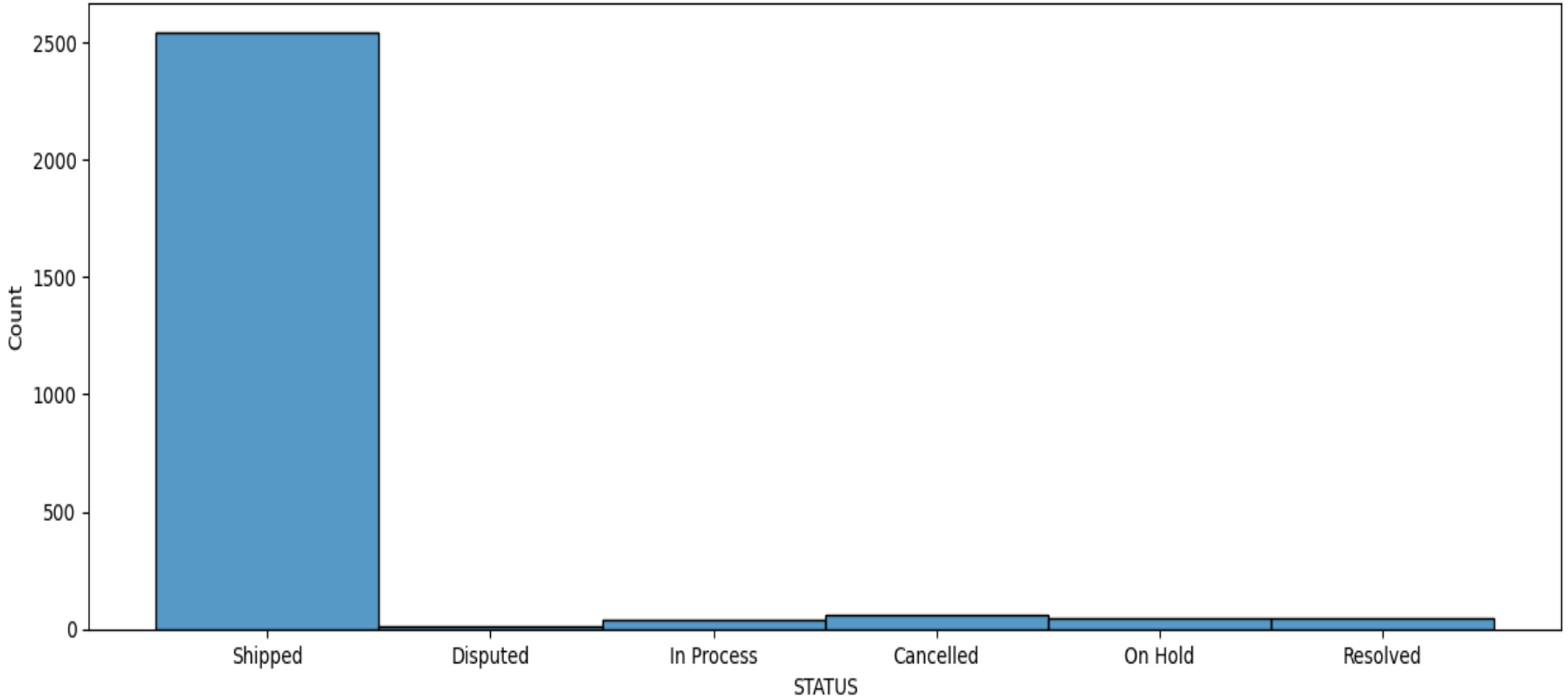




Bar graph for 'PRODUCTLINE'

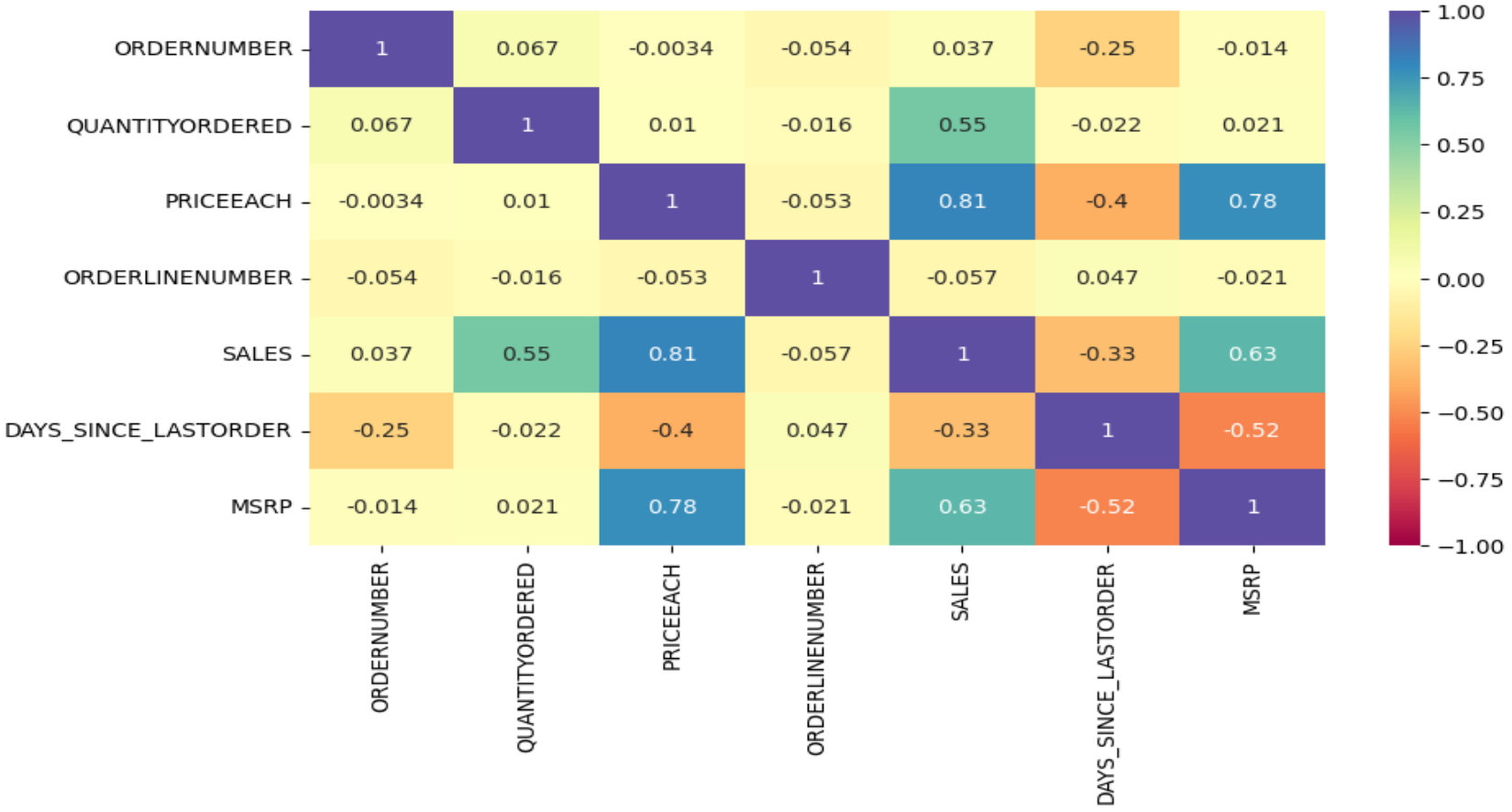


plot the histogram for the STATUS

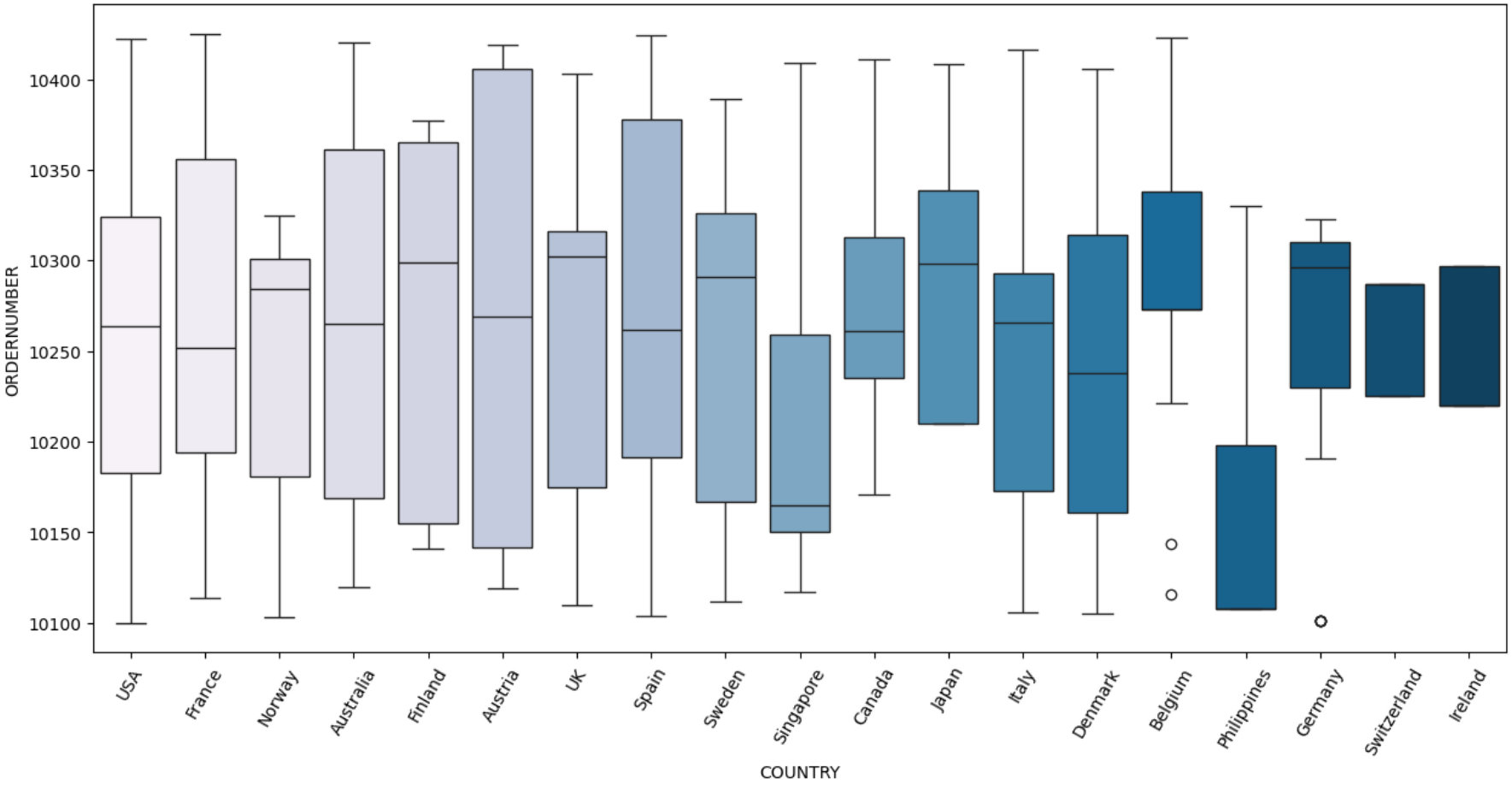


Bivariate Analysis

Sales and PRICEEACH have more correlation (0.81) where as for negative correlation (-0.52) MSRP & DAYS_SINCE_LASTORDER.
PRICEEACH and MSRP also have more correlation (0.78) second one.

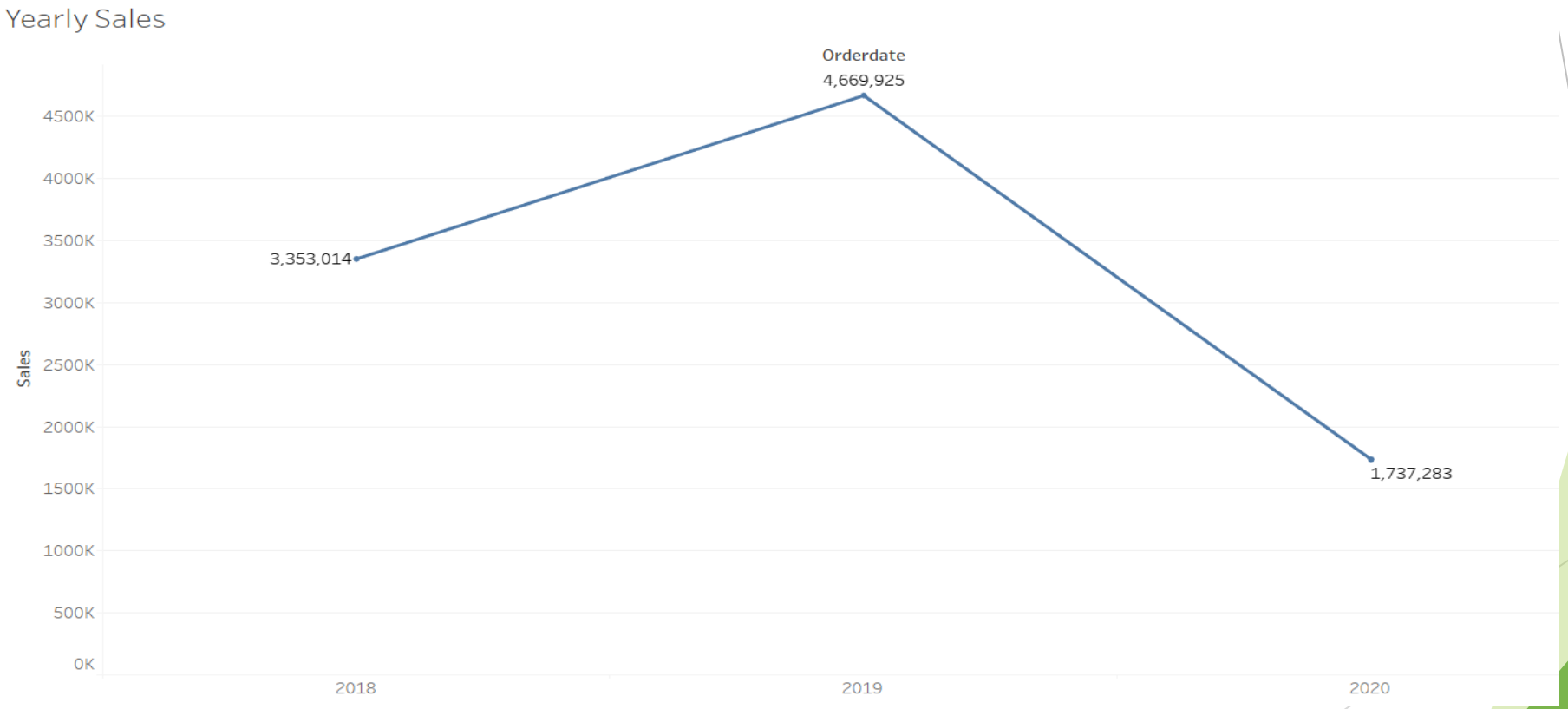


Multivariate Analysis
Relationship between country and ORDERNUMBER



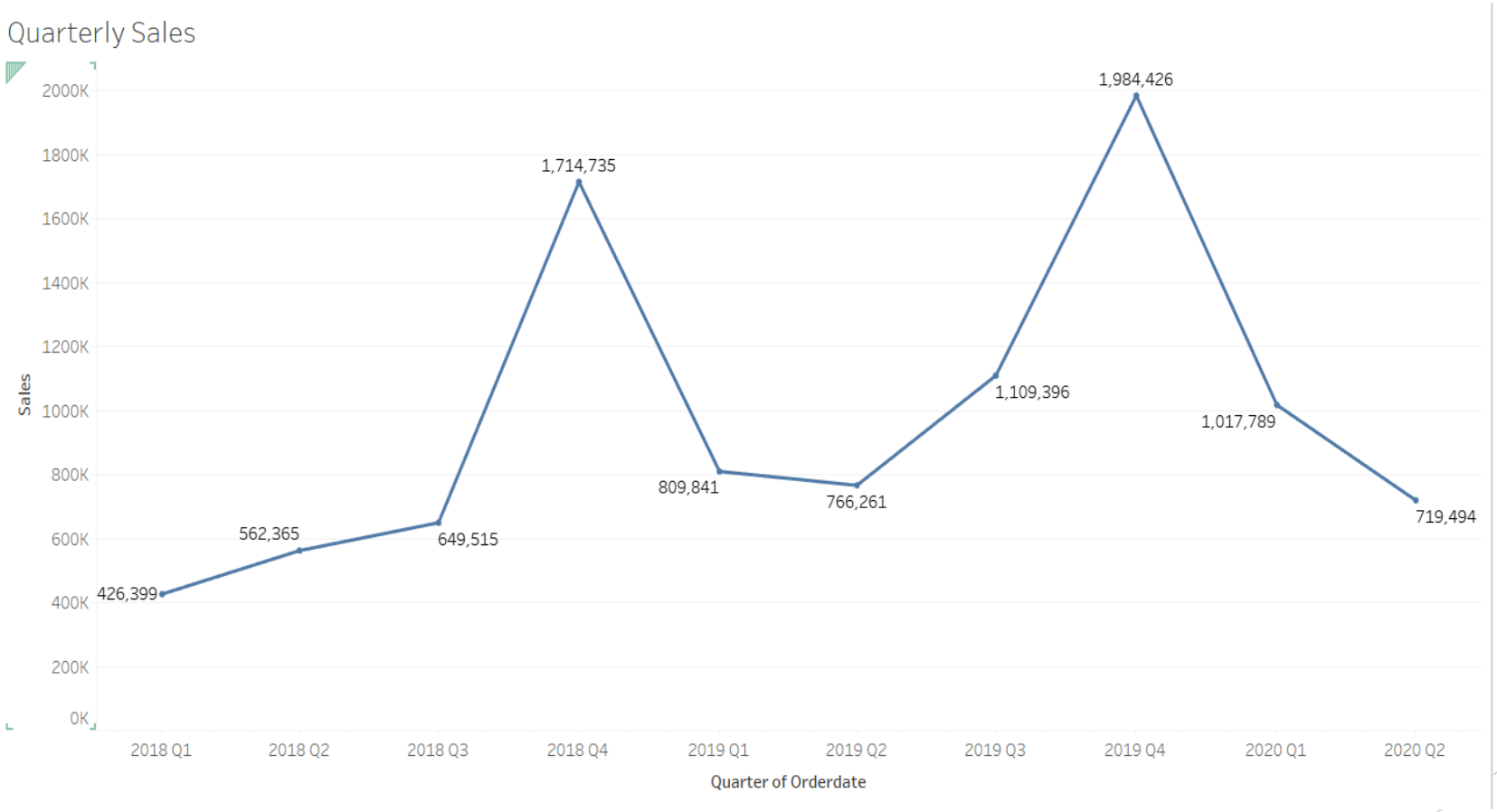
Yearly Sales

It booked the highest sale orders in 2019.



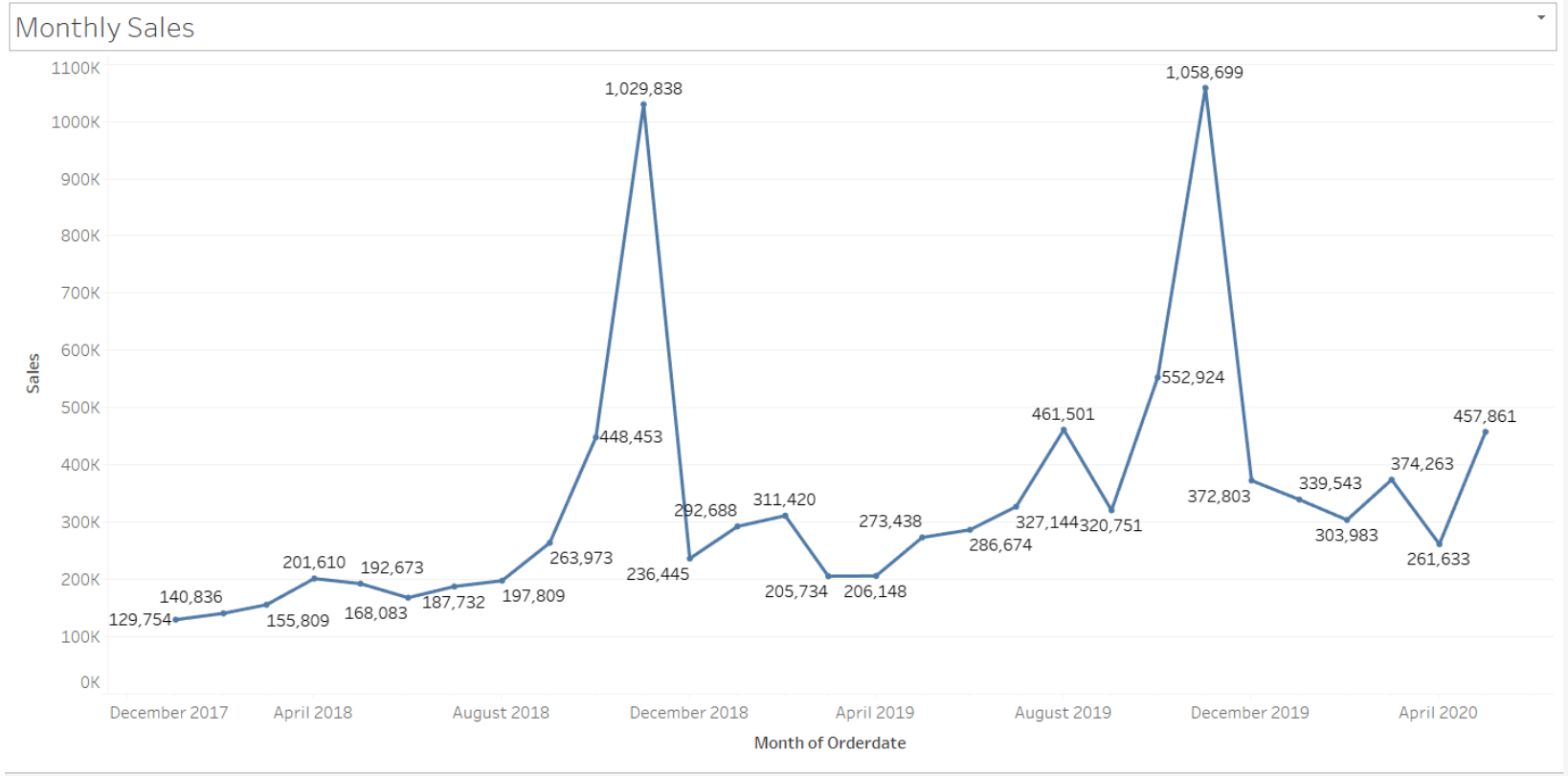
Quarterly Sales

Quarterly sales were highest in Q4 for 2018 and 2019.
It has lowest sales in Q1 of 2018, Q2 in 2019 & 2020.



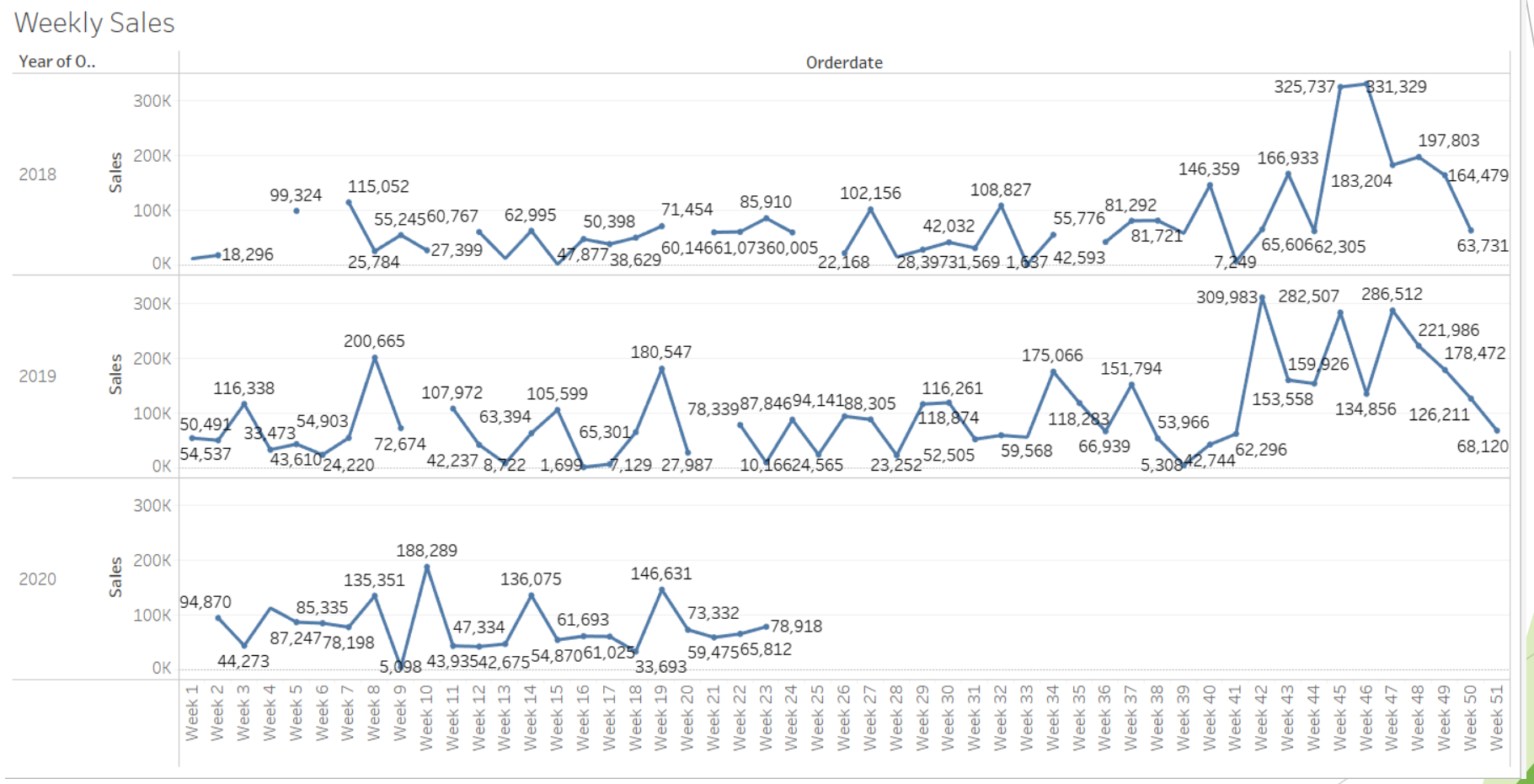
Monthly Sales

In December 2017, sales were very low.
During Oct – Nov months in 2018 & 2019, the sales are more.



Weekly Sales

It shows weekly sale orders.



Customer Segmentation using RFM analysis

RFM Analysis: -

RFM has been abbreviated as below

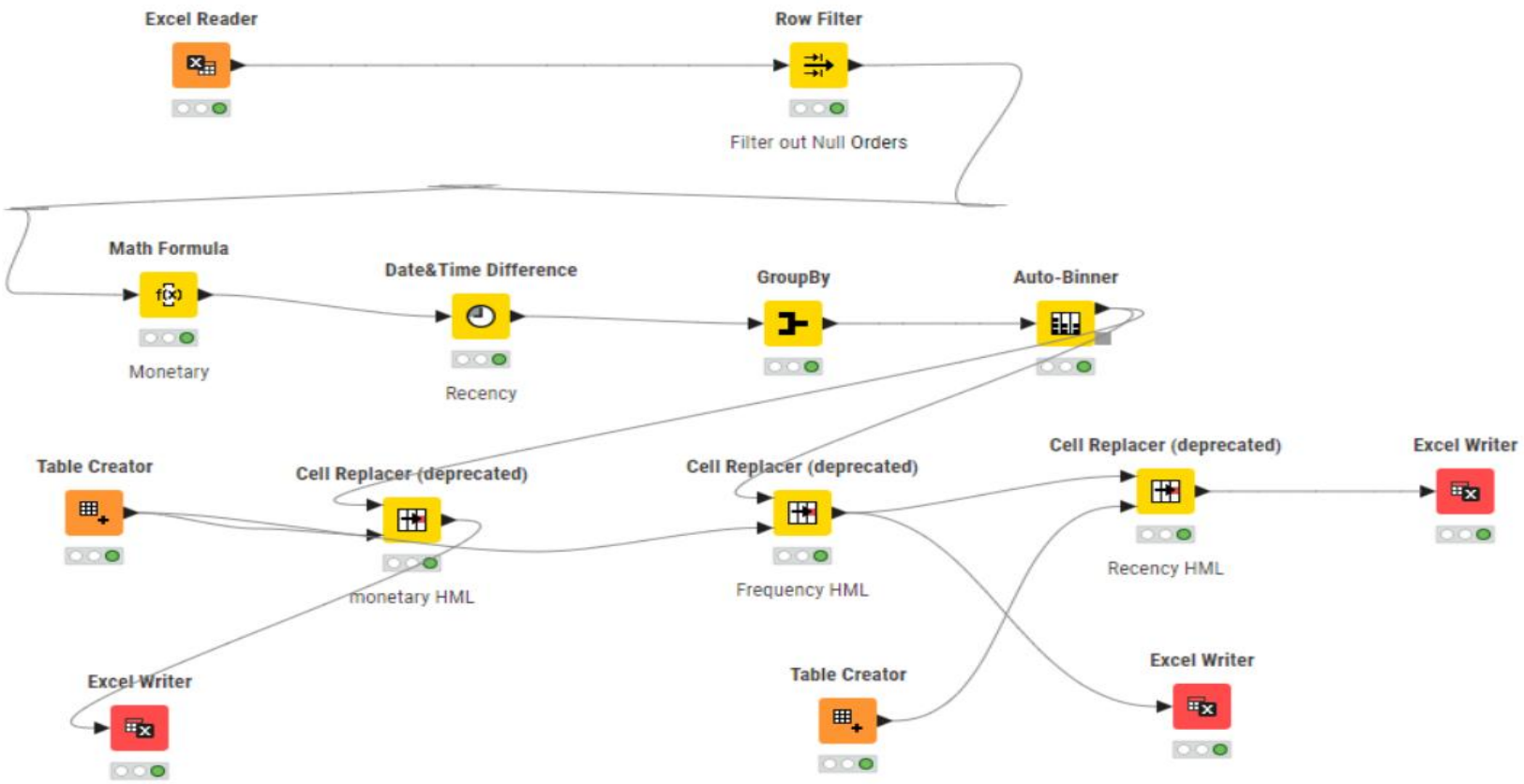
R – Recency – Recent/latest sales Transaction.

F – Frequency – No. of sale transactions.

- M – Monetary – Amount/Value for every sales transaction.

KNIME Workflow: -

KNIME Tool has used to analyse the Saleas_data and added the workflow screenshot.



Output details: -

This is a screen shot for RFM output.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA
1	ORDERNU	QUANTITY	PRICEEAC	ORDERLIN	SALES	ORDERDA	DAYS_SIN	STATUS	PRODUCT	MSRP	PRODUCT	PHONE	ADDRESSL	CITY	POSTALCC	COUNTRY	CONTACT	CONTACT	DEALSIZE	Monetary	Recency	ORDERNU	Monetary	Recency [	MONETAF	FREQUEN	RECENCY_HLM
2	10100	4	90.12	3	3033.31	4	1972.5	Shipped	Vintage C	90.75	4	4	2304 Long	4	4	4	Young	Valarie	Medium	12133.3	877	Bin 1	Bin 1	Bin 3	L	L	L
3	10101	4	95.3425	4	2858.09	4	2009.5	Shipped	Vintage C	93	4	4	Lyonerstr.	4	4	4	Keitel	Roland	Medium	11432.3	874	Bin 1	Bin 1	Bin 3	L	L	L
4	10102	2	86.715	2	3432.03	2	1339	Shipped	Vintage C	77.5	2	2	2678 King	2	2	2	Frick	Michael	Medium	6864.05	873	Bin 1	Bin 1	Bin 3	L	L	L
5	10103	16	102.641	11	3418.88	16	1988.63	Shipped	Classic Ca	105.938	16	16	Erling Skal	16	16	16	Bergulfser	Jonas	Medium	54702	854	Bin 1	Bin 3	Bin 3	H	L	L
6	10104	13	106.725	1	3432.46	13	2241.15	Shipped	Classic Ca	105	13	13	C/ Moralz	13	13	13	Freyre	Diego	Medium	44622	852	Bin 1	Bin 2	Bin 3	M	L	L
7	10105	15	107.317	2	3924.74	15	2458.87	Shipped	Classic Ca	107.267	15	15	Vinb'ltet 3	15	15	15	Petersen	Jytte	Large	58871.1	841	Bin 1	Bin 3	Bin 3	H	L	L
8	10106	18	85.0578	12	3121.18	18	2761.22	Shipped	Planes	84.7778	18	18	Via Ludovi	18	18	18	Rovelli	Giovanni	Medium	56181.3	835	Bin 1	Bin 3	Bin 3	H	L	L
9	10107	8	115.686	2	3222.97	8	1597	Shipped	Motorcycl	112.875	8	8	897 Long	8	8	8	Yu	Kwai	Small	25783.8	828	Bin 1	Bin 2	Bin 3	M	L	L
10	10108	16	96.57	6	3452.81	16	2097.69	Shipped	Classic Ca	99.4375	16	16	15 McCall	16	16	16	Cruz	Arnold	Medium	55245	821	Bin 1	Bin 3	Bin 3	H	L	L
11	10109	6	125.153	4	4566.47	6	1759.33	Shipped	Classic Ca	127	6	6	11328 Dou	6	6	6	Hernande	Rosa	Medium	27398.8	814	Bin 1	Bin 2	Bin 3	M	L	L
12	10110	16	88.335	16	3188.62	16	2091.38	Shipped	Classic Ca	90.75	16	16	Fauntleroy	16	16	16	Ashworth	Victoria	Medium	51017.9	806	Bin 1	Bin 3	Bin 3	H	L	L
13	10112	2	179.51	1	4874.5	2	1236.5	Shipped	Classic Ca	157.5	2	2	Berguvsvg	2	2	2	Berglund	Christina	Large	9749	800	Bin 1	Bin 1	Bin 3	L	L	L
14	10113	4	95.3275	2	3099.64	4	1824	Shipped	Trucks and	91.5	4	4	5677 Stror	4	4	4	Nelson	Valarie	Medium	12398.6	798	Bin 1	Bin 1	Bin 3	L	L	L
15	10114	10	105.579	8	3821.74	10	2169.8	Shipped	Classic Ca	103.2	10	10	265, boule	10	10	10	Bertrand	Marie	Medium	38217.4	792	Bin 1	Bin 2	Bin 3	M	L	L
16	10115	5	117.228	5	4955.48	5	2119.4	Shipped	Trucks and	114.2	5	5	5905 Pom	5	5	5	Hernande	Maria	Medium	24777.4	789	Bin 1	Bin 2	Bin 3	M	L	L
17	10116	1	63.38	1	1711.26	1	2963	Shipped	Trains	62	1	1	Boulevard	1	1	1	Cartrain	Pascale	Small	1711.26	782	Bin 1	Bin 1	Bin 3	L	L	L
18	10117	12	109.281	9	3638.12	12	2177.5	Shipped	Classic Ca	118	12	12	Bronz Sok.	12	12	12	Natividad	Eric	Medium	43657.5	777	Bin 1	Bin 2	Bin 3	M	L	L
19	10118	1	117.2	1	4219.2	1	3389	Shipped	Ships	100	1	1	Rambla de	1	1	1	Saavedra	Eduardo	Medium	4219.2	772	Bin 1	Bin 1	Bin 3	L	L	L
20	10119	14	84.5464	11	2759.22	14	2515.93	Shipped	Classic Ca	86.2143	14	14	Geislweg	14	14	14	Pipps	Georg	Medium	38629.1	765	Bin 1	Bin 2	Bin 3	M	L	L
21	10120	15	97.1553	3	3359.84	15	2413.93	Shipped	Motorcycl	94.8	15	15	636 St Kilc	15	15	15	Ferguson	Peter	Small	50397.7	764	Bin 1	Bin 3	Bin 3	H	L	L
22	10121	5	97.1	5	3794.39	5	2047.6	Shipped	Motorcycl	99.4	5	5	59 rue de	5	5	5	Henriot	Paul	Small	18972	756	Bin 1	Bin 2	Bin 3	M	L	L

Parameters: -

Recency has calculated by this formula Maximum order date (31.05.2020) – Order date
Frequency is count of customer names.
Monetary is customer wise sum (PRICEEACH * QUANTITYORDERED).

▶ Inferences from RFM Analysis: -

- Top five Customers

Customer Name	MONETARY_HLM	FREQUENCY_HLM	RECENCY_HLM
Ferguson Peter	H	H	H
Freyre Diego	H	H	H
Frick Sue	H	H	H
Henriot Paul	H	H	H
Koskitalo Pirkko	H	H	H
Labrune Janine	H	H	H
Nelson Valarie	H	H	H
Pipps Georg	H	H	H
Suominen Kalle	H	H	H
Yoshido Juri	H	H	H

Best/Loyal customers

These are the best customers because they have visited recently with high monetary contributions. Somehow interested to visit frequently.

Customer Name	MONETARY_HLM	FREQUENCY_HLM	RECENCY_HLM
Accorti Paolo	H	M	M
Ashworth Victoria	H	M	M
Barajas Miguel	H	M	M
Benitez Violeta	H	M	M
Bennett Helen	H	M	M
Bertrand Marie	H	M	M
Devon Elizabeth	H	M	M
Dewey Catherine	H	M	M
Ferguson Peter	H	M	M
Franco Valarie	H	M	M
Freyre Diego	H	M	M
Frick Michael	H	M	M

Customers on the verge of churning: -

These customers are dissatisfied and searching for other vendors for better service. So, they not coming to visit the shop recently. Hence, it needs to improve the areas wherever it can control the attrition rate.

Customer Name	MONETARY_HLM	FREQUENCY_HLM	RECENCY_HLM
Devon Elizabeth	M	M	L
Keitel Roland	M	M	L
Klaeboe Jan	M	M	L

Lost customers

The above list is related to lost customers because they have not preferred to come frequently and are not interested in buying anything. So it is required to offer them with high benefit products to the customers to improve sales. It reduces the attrition rate of the customers.

Customer Name	MONETARY_HLM	FREQUENCY_HLM	RECENCY_HLM
Benitez Violeta	L	L	L
Berglund Christina	L	L	L
Bergulfsen Jonas	L	L	L
Calaghan Tony	L	L	L
Cartrain Pascale	L	L	L
Ferguson Peter	L	L	L
Fresnisre Jean	L	L	L
Freyre Diego	L	L	L
Frick Michael	L	L	L
Henriot Paul	L	L	L
Keitel Roland	L	L	L
Lewis Dan	L	L	L
Mendel Roland	L	L	L
Nelson Valarie	L	L	L
Roulet Annette	L	L	L
Saavedra Eduardo	L	L	L
Schmitt Carine	L	L	L
Taylor Leslie	L	L	L
Young Leslie	L	L	L
Young Valarie	L	L	L

Inferences and Recommendations: -

Inferences: -

- Yearly sales have been reduced, so it is not a good sign.
- Many of the orders from the USA and UK occupied a lesser market.
- Month-end sales are much less compared to other days.
- Quarterly sales are more for Q4 in 2018 & 2019.

Recommendation:-

- Sales need to improve and focus on marketing.
- The market needs to expand to other countries as well.
- It is required to put some offers.
- It needs to focus on other than Q4 sales for better improvements to the customers.

Grocery retail industry

Contents: -

- Problem Statement (PART B) and Exploratory Data Analysis
- Market Basket Analysis
- Inferences and Recommendations

Problem Statement (Part B)

Business Context: -

- In the highly competitive grocery retail industry, understanding customer buying patterns is crucial for enhancing sales, increasing customer satisfaction, and improving profitability. By identifying frequently purchased item combinations, grocery stores can craft effective marketing strategies, optimize inventory management, and tailor promotions to meet customer needs. Leveraging Point of Sale (POS) data can unlock valuable insights that drive customer-centric offerings, such as combo packs, discounts, and targeted promotions, which can increase basket size and improve customer retention. This analysis aligns with business goals by maximizing revenue, reducing operational costs, and boosting customer loyalty.

Objective: -

- As a business analyst, the goal is to analyze the POS transactional data to identify frequently purchased item combinations. Using association rule mining or similar techniques, the aim is to uncover patterns that will help the store create targeted combo offers and discounts, ultimately driving revenue growth by increasing customer purchases and average basket size.

Data Description:

The dataset consists of transactional data from a grocery store, where each row represents a product purchased in a specific order. The columns in the dataset are as follows:

- **Date:** The date when the transaction took place.
- **Order_id:** A unique identifier for each customer order.
- **Product:** The individual item purchased in the transaction.

PART B: Define the Problem and conduct an Exploratory Data Analysis

Univariate Analysis

These are the unique values of the various attributes.

Unique values of Order_id

1139

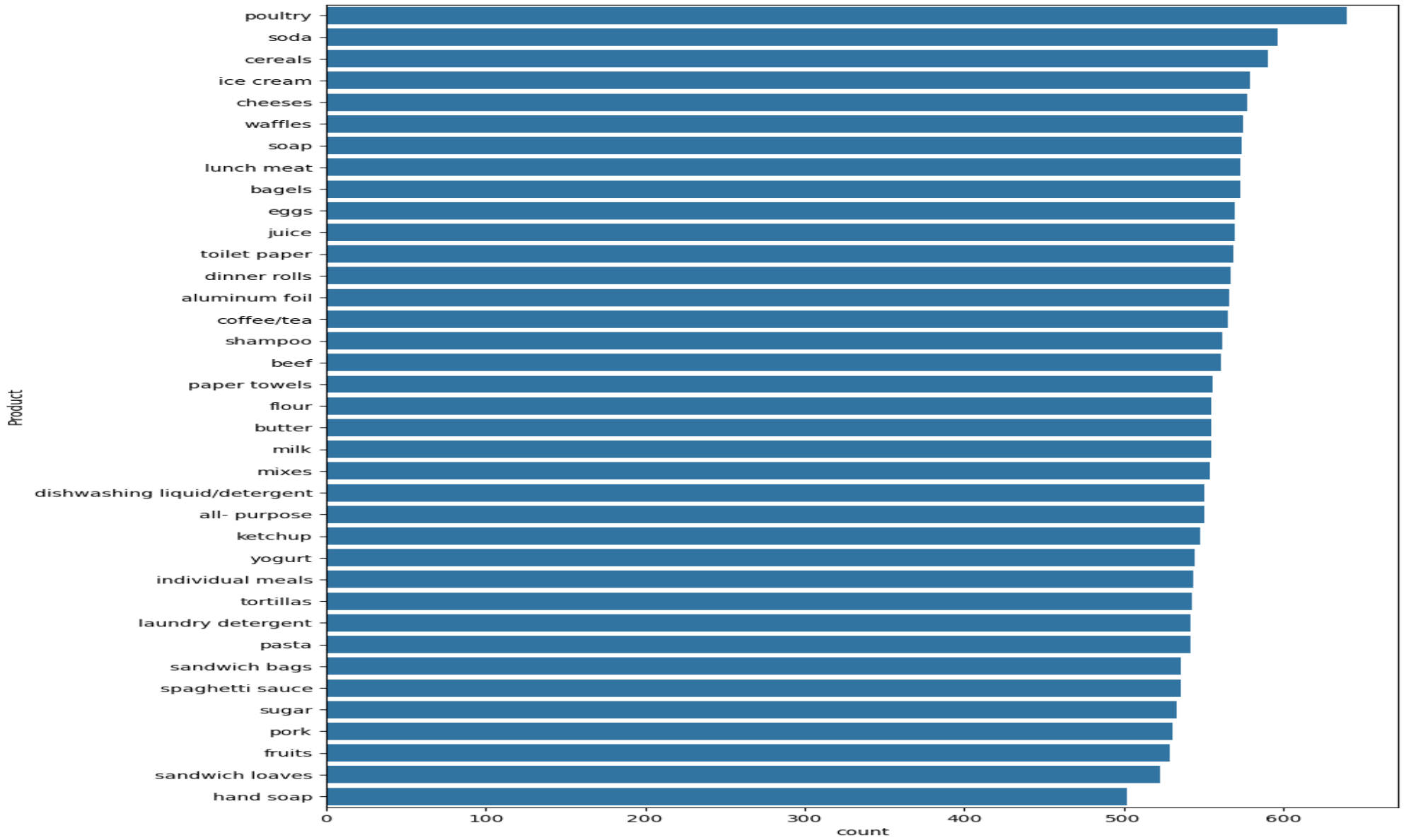
10.2 Unique values of Date

603

Unique values of Product

37

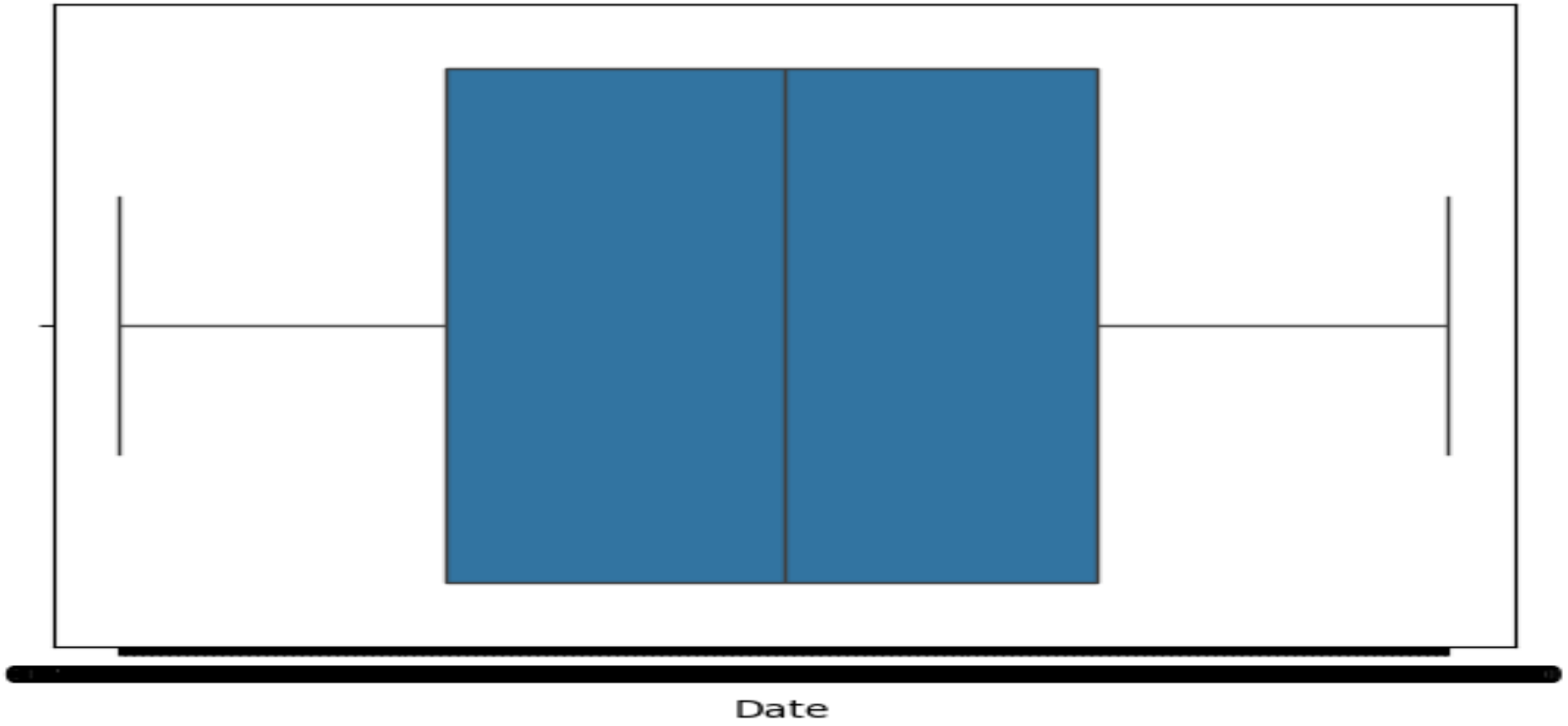
Product


poultry
soda
cereals
ice cream
cheeses
waffles
soap
lunch meat
bagels
eggs
juice
toilet paper
dinner rolls
aluminum foil
coffee/tea
shampoo
beef
paper towels
flour
butter
milk
mixes
dishwashing liquid/detergent
all- purpose
ketchup
yogurt
individual meals
tortillas
laundry detergent
pasta
sandwich bags
spaghetti sauce
sugar
pork
fruits
sandwich loaves
hand soap
Product
0
100
200
300
400
500
600
count

- Poultry product has the highest number of orders as per the provided dataset.
- Soda is in second place with more orders where as the cereals in third place.
- Butter, milk, and flour consist number of orders i.e 555.
- Handsoap has a fewer number of orders.

Date

There are no outliers in the boxplot, but left hand side is more comparatively right-hand side.



Bivariate Analysis

It is a relationship of numeric values, but the given dataset has only order_id with numeric values. Hence, the heatmap considered the relationship with order_id only. Overall, there is no relationship between any two variables.

Multivariate analysis: -
Date, Product & Order_id

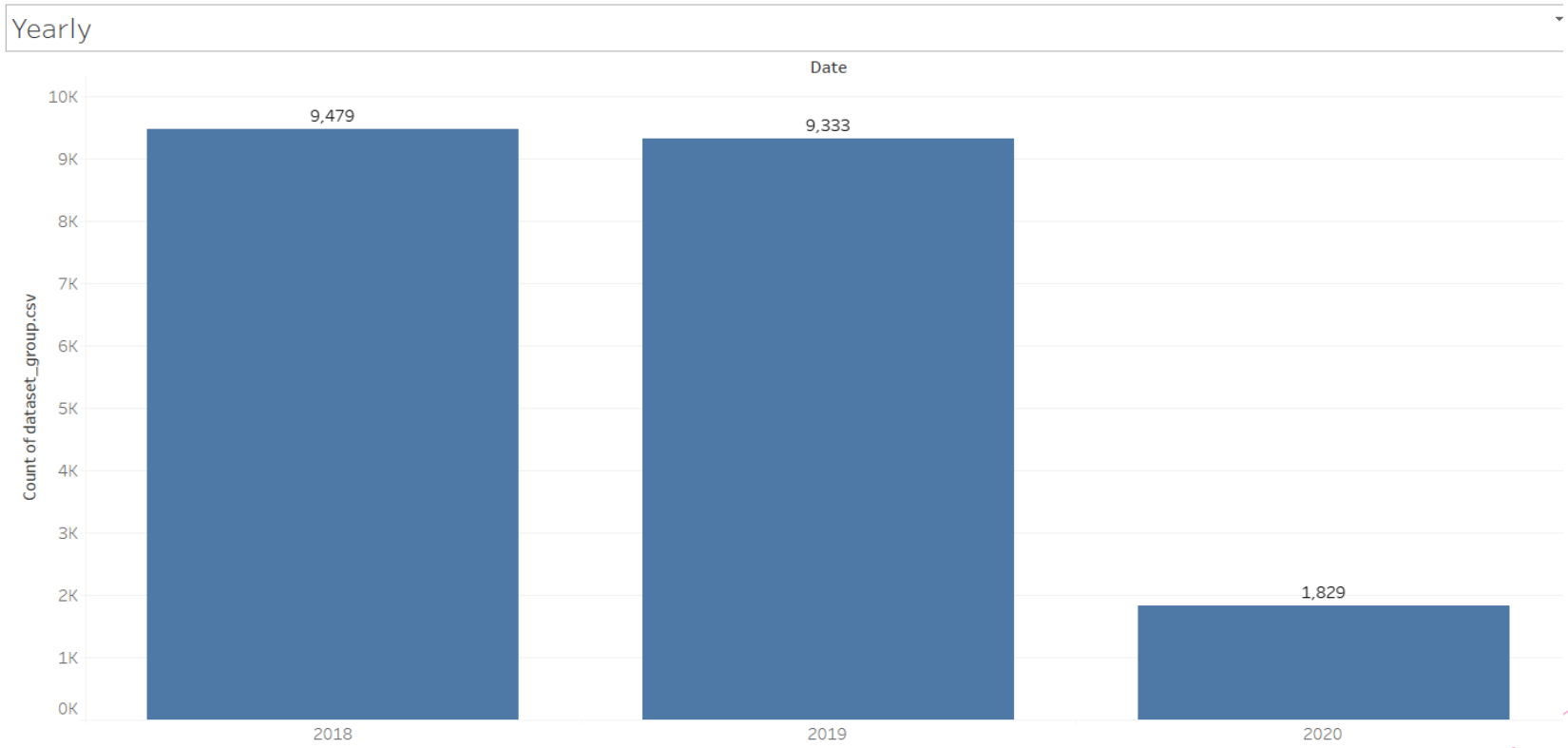
The below table shows the date, product, and order_id.

Date	Product	Order_id
08-02-2019	paper towels	11
20-02-2019	soda	10
17-05-2018	soda	9
01-03-2018	waffles	9
08-02-2019	mixes	9
20-02-2019	spaghetti sauce	9
02-08-2018	cereals	9
20-02-2019	soap	8
13-08-2019	lunch meat	8
09-01-2019	sandwich loaves	8
06-03-2018	coffee/tea	8
08-02-2019	milk	8
15-07-2018	flour	8
03-02-2018	hand soap	8

dtype: int64

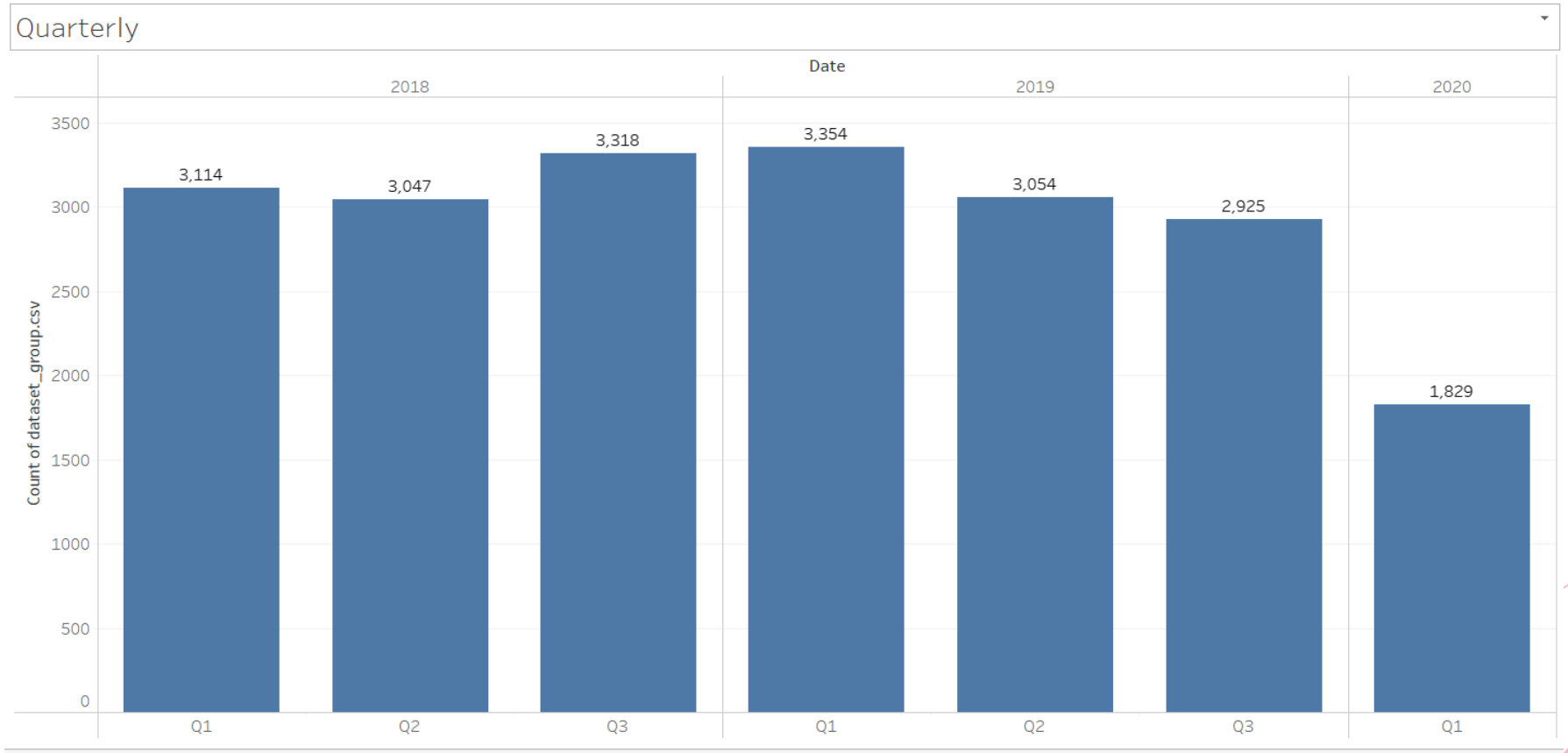
Yearly sales

This chart shows the yearly sales. In the year, 2018, it had more sales, 9,479, whereas it was reduced to 1,829 in 2020.



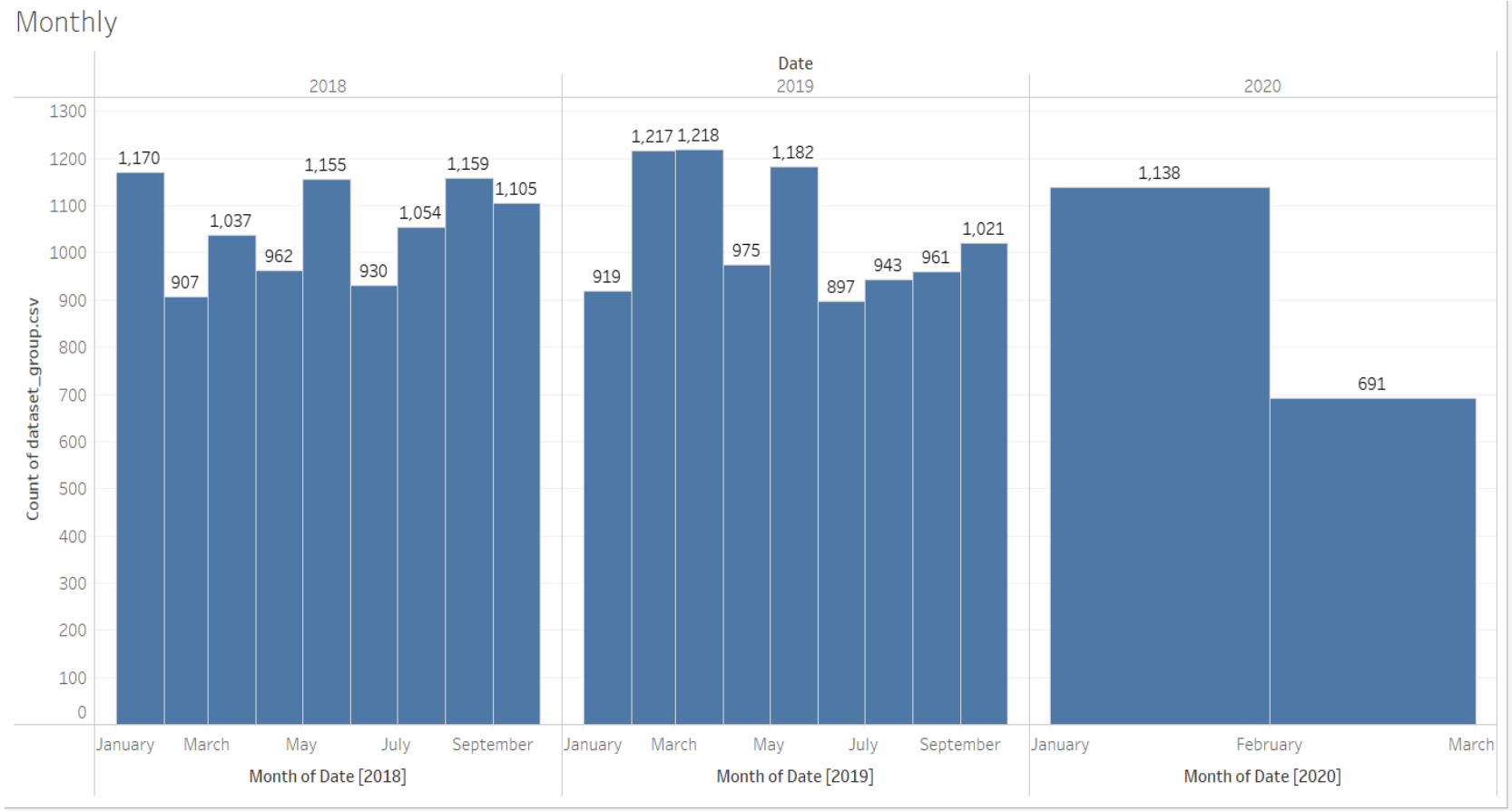
Quarterly Sales

The bar chart represents the sales of different quarters in different years.
2018 – Q3 has highst results comparatively Q2 & Q3.
2019 – Highest sales in Q1 and lowest sales in Q3.
2020 – Shows the sales relating to Q1 only.



Monthly Sales

The histogram shows the month-wise sales in each year.

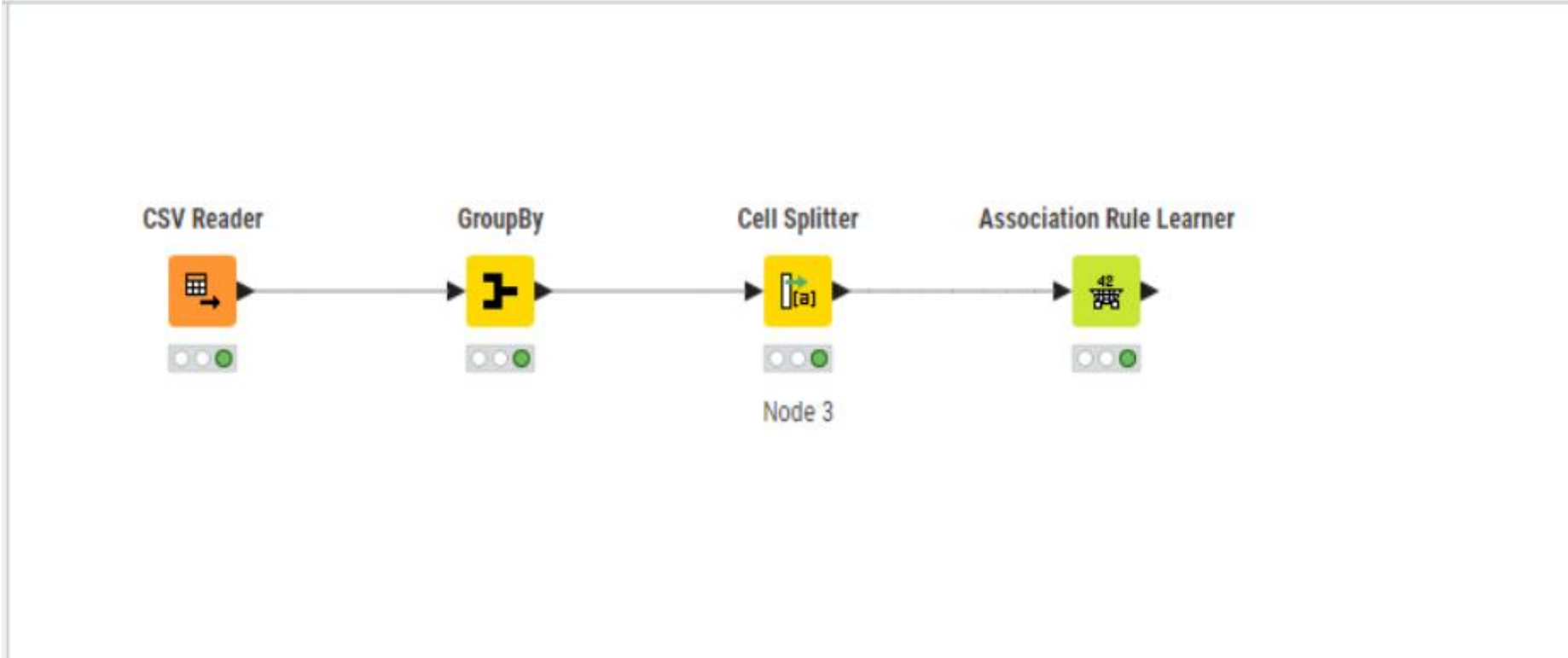


Market Basket Analysis

- It is a Data mining technique used to analyse the data.
- It depends on the Frequency, monetary and Recency of the customers of a business.
- The given data set relates to Grocery products to analyse.

KNIME Workflow: -

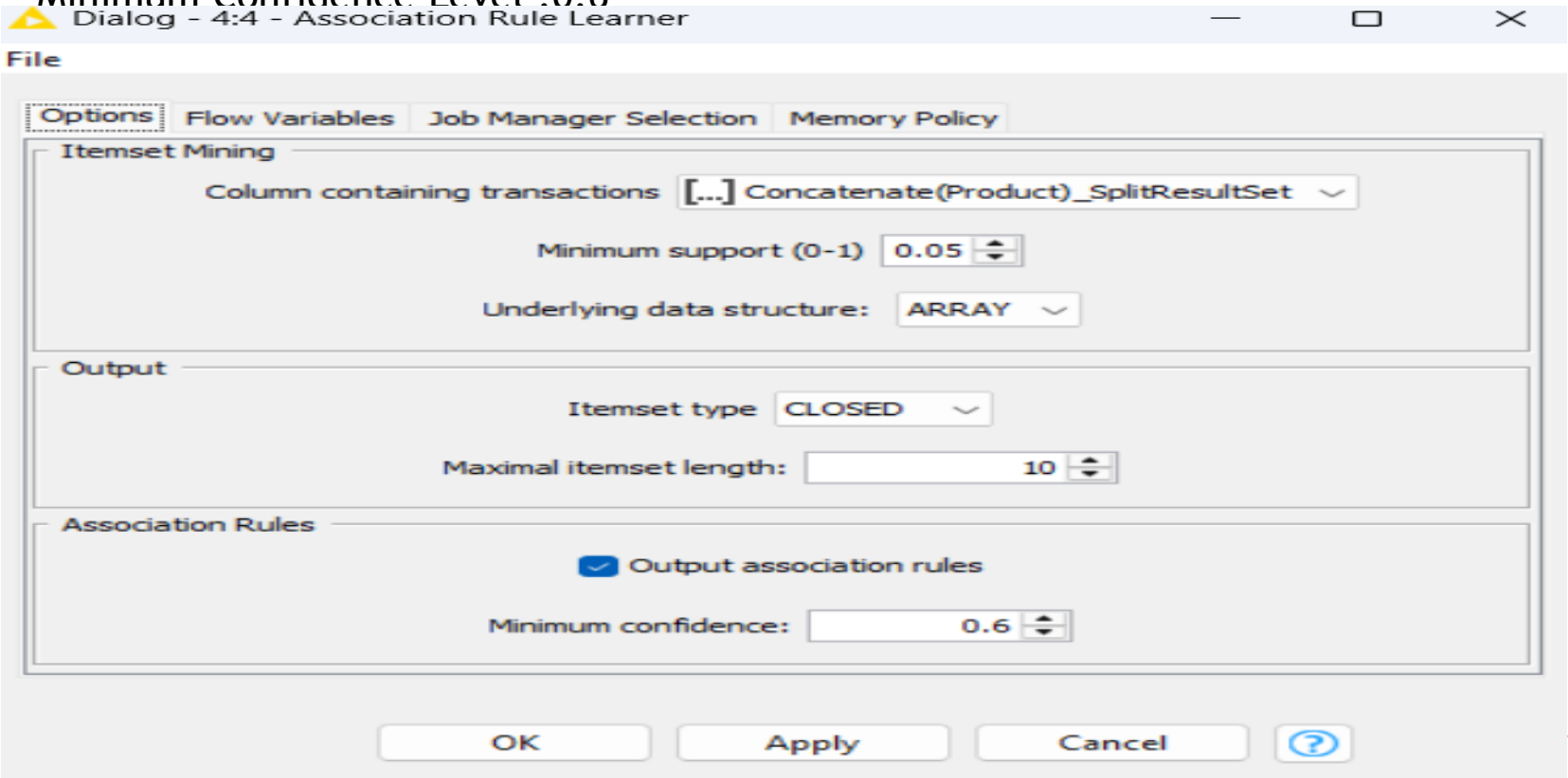
KNIME workflow for Market Basket Analysis for the provided data.



Parameters:-

The below parameters are used for the Market-Basket analysis output table (Association Rule Learner).

Minimum Support: 0.05
Maximum Item Set Length: 10
Minimum Confidence Level :0.6



Output table:-

Frequent itemsets/Association rules (Table)

Rows: 24 | Columns: 6

#	RowID	Support Number (double)	Confidence Number (double)	Lift Number (double)	Consequent String	implies String	Items Set
1	rule0	0.05	0.64	1.7	juice	<---	[yogurt,toilet paper,aluminum foil]
2	rule1	0.05	0.62	1.645	juice	<---	[yogurt,poultry,aluminum foil]
3	rule2	0.05	0.613	1.616	coffee/tea	<---	[yogurt,cheeses,cereals]
4	rule3	0.05	0.6	1.424	poultry	<---	[dishwashing liquid/detergent,laundry
5	rule4	0.051	0.63	1.678	mixes	<---	[yogurt,poultry,aluminum foil]
6	rule5	0.051	0.611	1.66	sandwich bags	<---	[cheeses,bagels,cereals]
7	rule6	0.051	0.674	1.726	cheeses	<---	[bagels,cereals,sandwich bags]
8	rule7	0.051	0.617	1.558	cereals	<---	[cheeses,bagels,sandwich bags]
9	rule8	0.051	0.63	1.621	dinner rolls	<---	[spaghetti sauce,poultry,cereals]
10	rule9	0.051	0.637	1.512	poultry	<---	[dinner rolls,spaghetti sauce,cereals]
11	rule10	0.051	0.604	1.589	milk	<---	[poultry,laundry detergent,cereals]
12	rule11	0.052	0.628	1.61	eggs	<---	[dinner rolls,poultry,soda]
13	rule12	0.052	0.641	1.649	dinner rolls	<---	[spaghetti sauce,poultry,ice cream]
14	rule13	0.052	0.686	1.628	poultry	<---	[dinner rolls,spaghetti sauce,ice crear
15	rule14	0.052	0.628	1.614	dinner rolls	<---	[spaghetti sauce,poultry,juice]
16	rule15	0.052	0.602	1.429	poultry	<---	[dinner rolls,spaghetti sauce,juice]
17	rule16	0.052	0.634	1.627	eggs	<---	[paper towels,dinner rolls,pasta]
18	rule17	0.052	0.602	1.621	pasta	<---	[paper towels,eggs,dinner rolls]
19	rule18	0.054	0.642	1.651	dinner rolls	<---	[spaghetti sauce,poultry,laundry deter
20	rule19	0.054	0.656	1.556	poultry	<---	[dinner rolls,spaghetti sauce,laundry (
21	rule20	0.055	0.624	1.565	ice cream	<---	[paper towels,eggs,pasta]
22	rule21	0.055	0.63	1.616	eggs	<---	[paper towels,ice cream,pasta]
23	rule22	0.055	0.643	1.731	pasta	<---	[paper towels,eggs,ice cream]
24	rule23	0.055	0.649	1.791	paper towels	<---	[eggs,ice cream,pasta]

Inferences and Recommendations: -

Inferences

- Some products have more than two recommendations, like yogurt, poultry and paper, towels, etc.
- Customers also prefer to buy a consequent product along with a set.
- It has involved high cost for few buckets/set.
- Customers will buy only high lift value products due to set and consequent combination.

Recommendations

- Higher lift will be recommended for the products with more than two recommendations. Ex. For papers, the towels set will recommend (papers, towels, eggs, and ice cream) which has the highest lift as 1.731. It is required to club helpful products to the customers.
- High lift value is preferable to add along with a set of products. Better to announce more combo packs or 1+1 offers to attract the customers.
- It is required to focus more on Lesser lift valued products with more offers or combo packs.